SEMI-SUPERVISED LEARNING FOR MESOTHELIOMA CLASSIFICATION

FP-01
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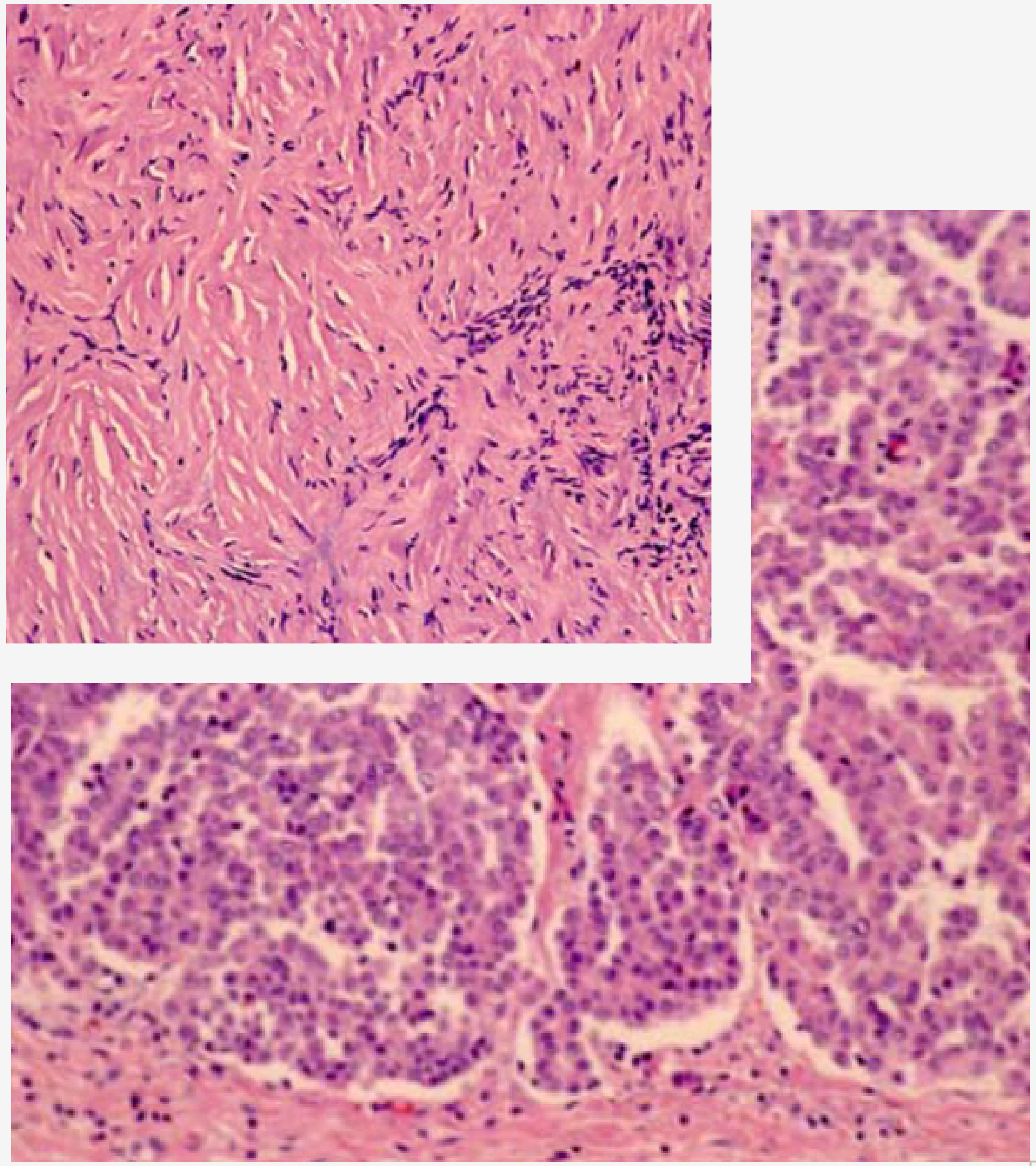
Mesothelioma

Malignant pleural mesothelioma is a **rare**, highly **aggressive cancer** originating from the mesothelial cells of the pleura (the lining around the lungs).
Its main cause is **asbestos exposure**, often with a latency period of several decades.

Main Aspect:

- Aggressiveness and Rarity
- Diagnostic Challenges
  - Early **symptoms** are non-specific.
  - Most cases are diagnosed at an **advanced stage**.
  - This **late diagnosis** leads to a poor prognosis and limited treatment options.
- Importance of Histological Classification



Pathology of mesothelioma – Inai, Environmental Health and Preventive Medicine (2008)
Diagnosis of malignant pleural mesothelioma – Alì et al., Translational Lung Cancer Research (2021)
Histologic subtyping of mesothelioma – Brcic & Kern, Translational Lung Cancer Research (2018)

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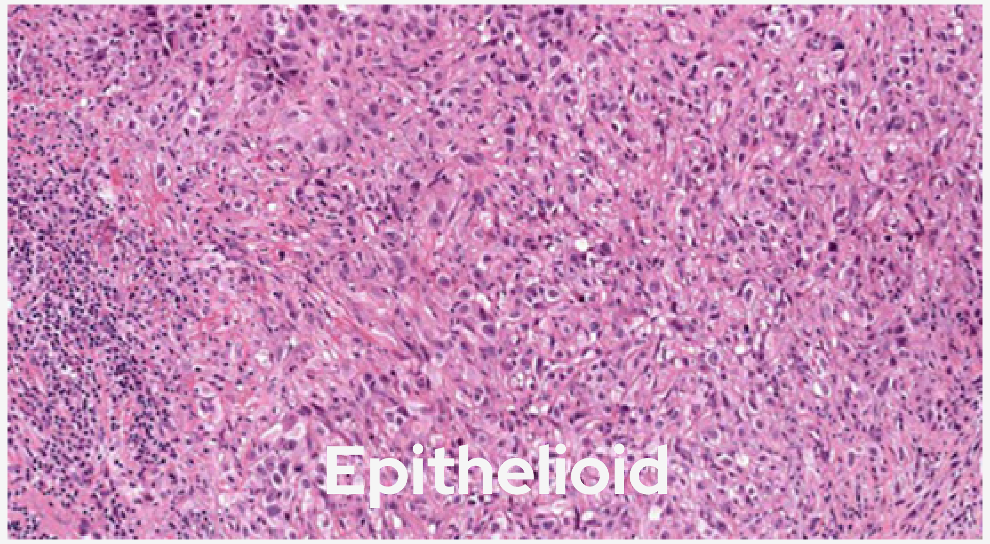
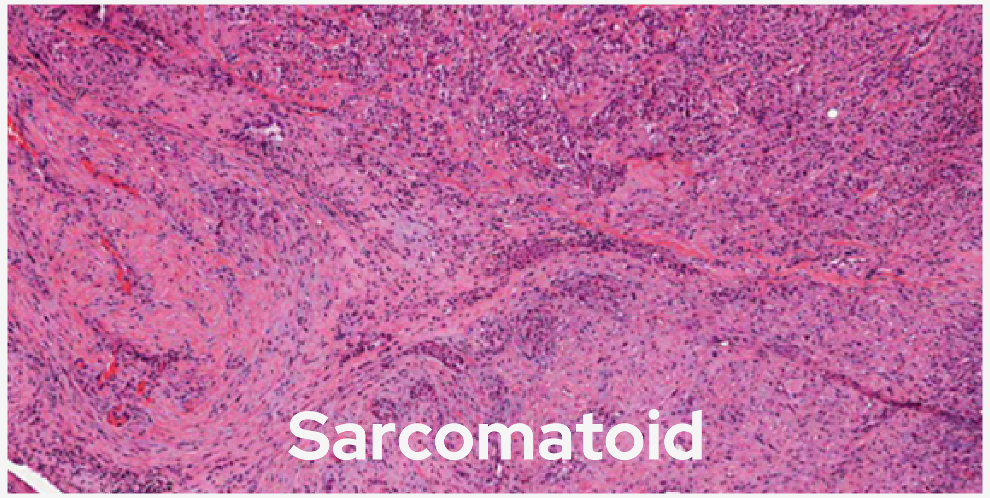
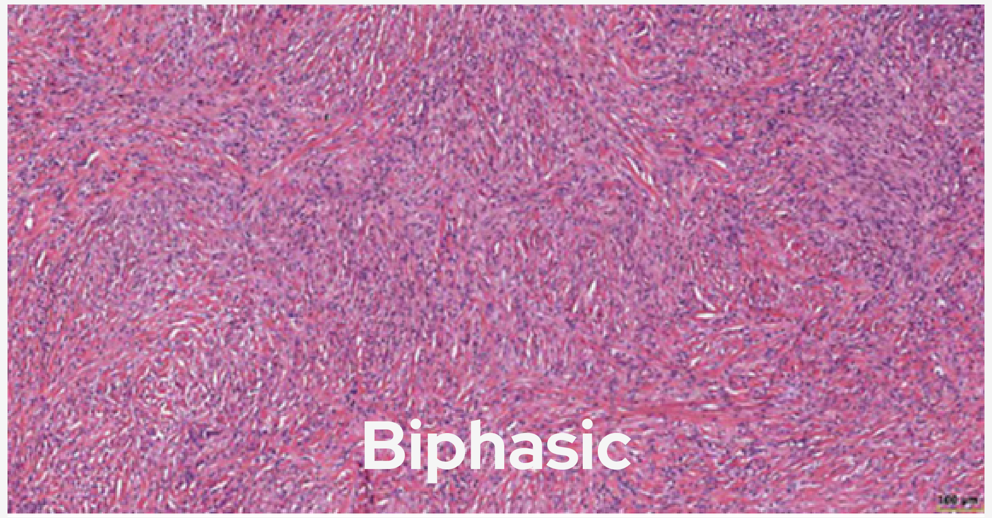
Histological Subtypes

- **Epithelioid (~60%)**
  - Composed of cells resembling normal mesothelial cells, often forming **papillary or tubular structures**
  - Best prognosis among the subtypes
- **Sarcomatoid (~20%)**
  - Characterized by **spindle-shaped cells** similar to those found in true sarcomas
  - Most aggressive subtype
- **Biphasic (~20%)**
  - Contains **both epithelioid and sarcomatoid** components within the same tumor







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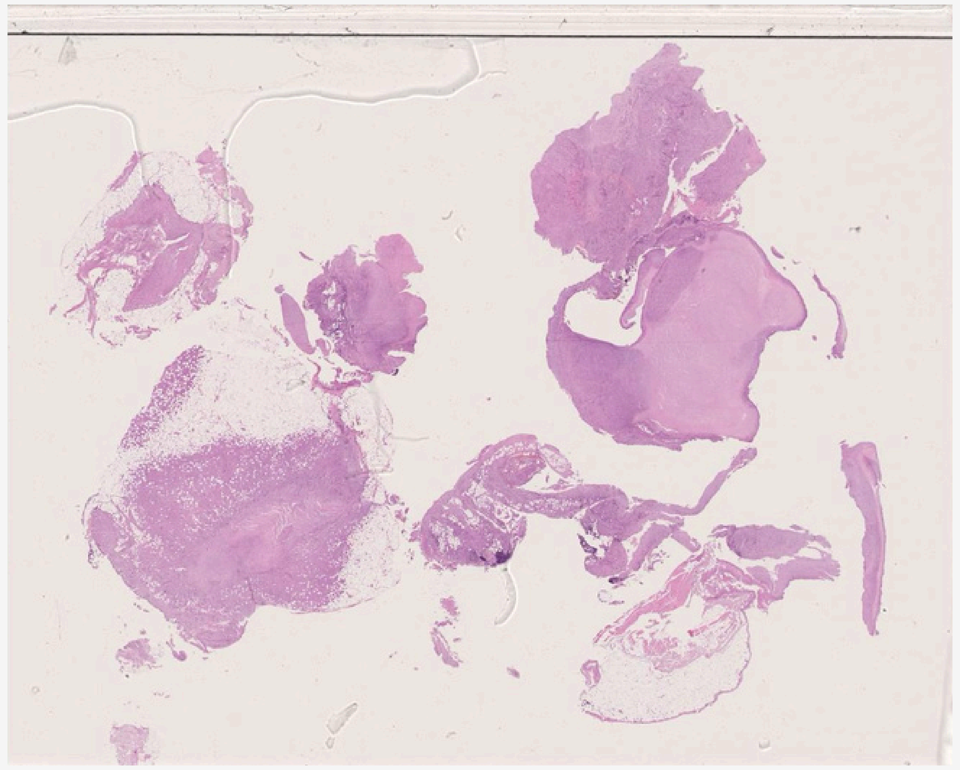

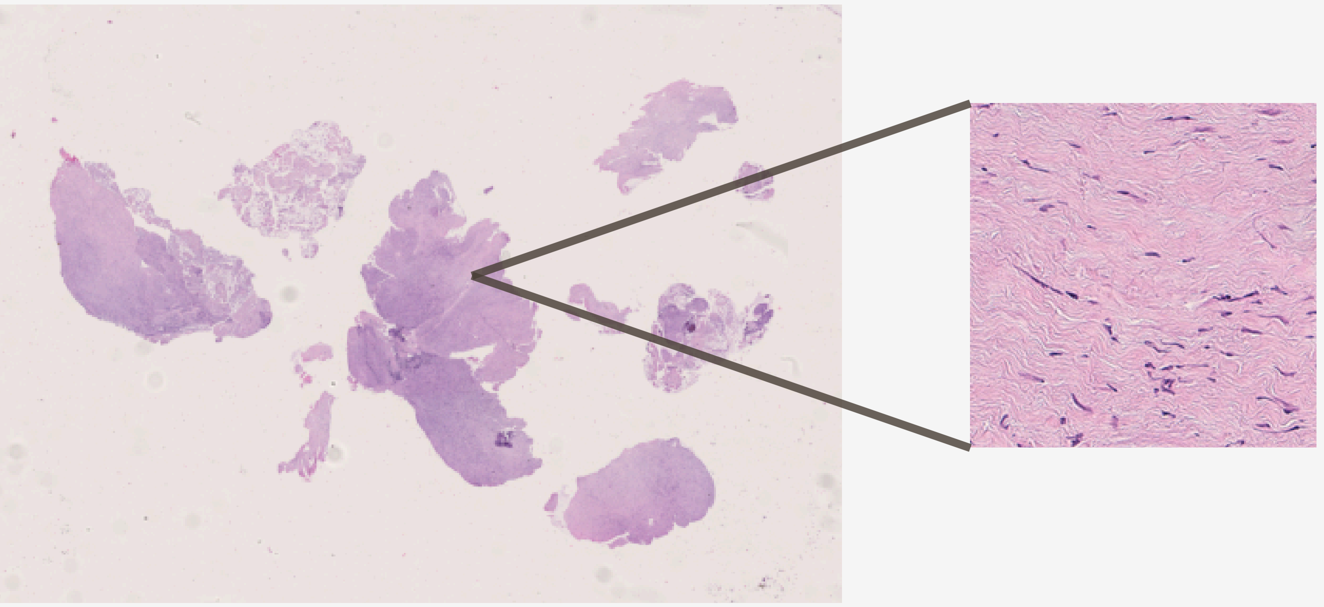

Technical Challenges in Whole Slide Images (WSI)

Whole Slide Images (WSI) have revolutionized digital pathology by enabling the acquisition and analysis of ultra-high-resolution images of entire tissue slides.

Technical challenges:

- Gigapixel Digital Images:
  - WSIs are extremely large, often millions to billions of pixels. This high resolution is essential for capturing fine histological details.
- High Computational Complexity:
  - Processing, storing, and analyzing WSIs require substantial computational resources.
- Limited and Costly Annotations:
  - Annotating WSIs is a time-consuming and expensive process, typically requiring expert pathologists.

Multiple instance learning in digital pathology – Gadermayr & Tschuchnig, IEEE Transactions on Medical Imaging (2022)

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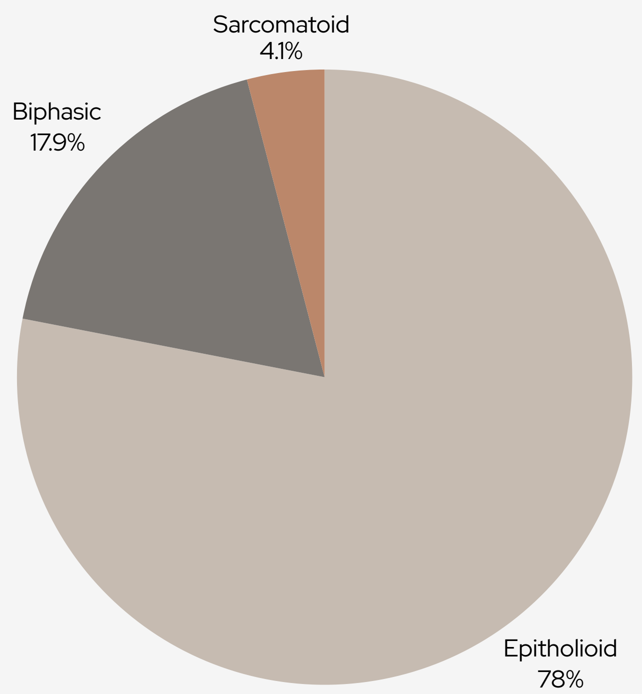
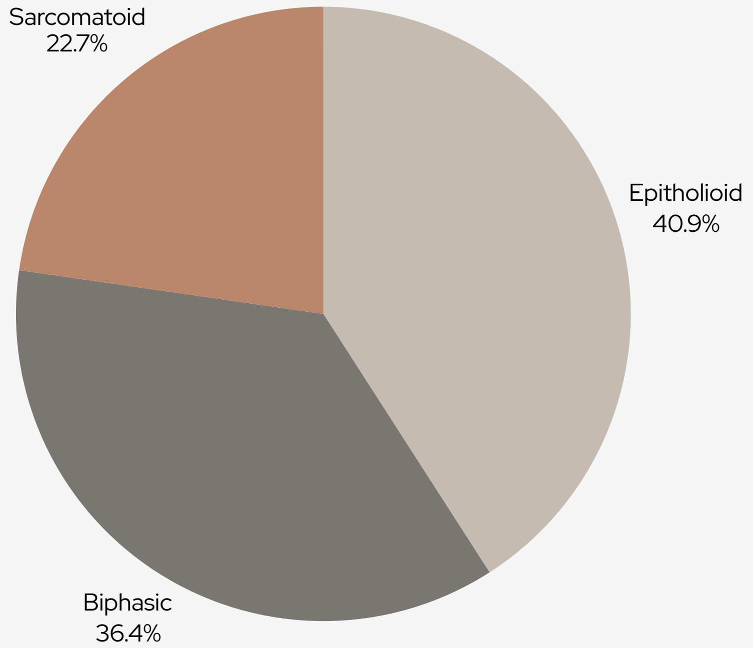
The Dataset





"Mesothelioma San Luigi"

123 cases included, covering all three major histological subtypes, granted by San Luigi Hospital

- Epithelioid: 96 cases
- Biphasic: 22 cases
- Sarcomatoid: 5 cases

Our Sub-Dataset

22 Whole Slide Images (WSI) selected for experiments
Dataset reflects real-world clinical prevalence and challenges

- Epithelioid: 9 cases
- Biphasic: 8 cases
- Sarcomatoid: 5 cases

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Methodological Approach (Part 1)

Semi-Supervised and Self-Supervised Learning in Digital Pathology

Semi-Supervised Learning

- Combines a **small amount of labeled** data with a **large amount of unlabeled** data to train models.
- Especially useful in pathology, where expert annotation is time-consuming and expensive.
- Allows models to leverage the abundance of available, unlabeled whole slide images (WSI) for improved performance.
- Reduces the annotation burden while maintaining competitive accuracy in classification tasks.

Self-Supervised Learning

- Models learn m**eaningful representations from unlabeled** data by solving pretext tasks.
- No external labels are required; the data itself provides the supervision signal.
- Particularly effective for extracting robust features in medical imaging, where labeled data is scarce.
- Enables better generalization and transferability to downstream tasks, such as WSI classification.

Semi-/self-supervised learning in histopathology – Voigt et al., Computers in Biology and Medicine (2022)

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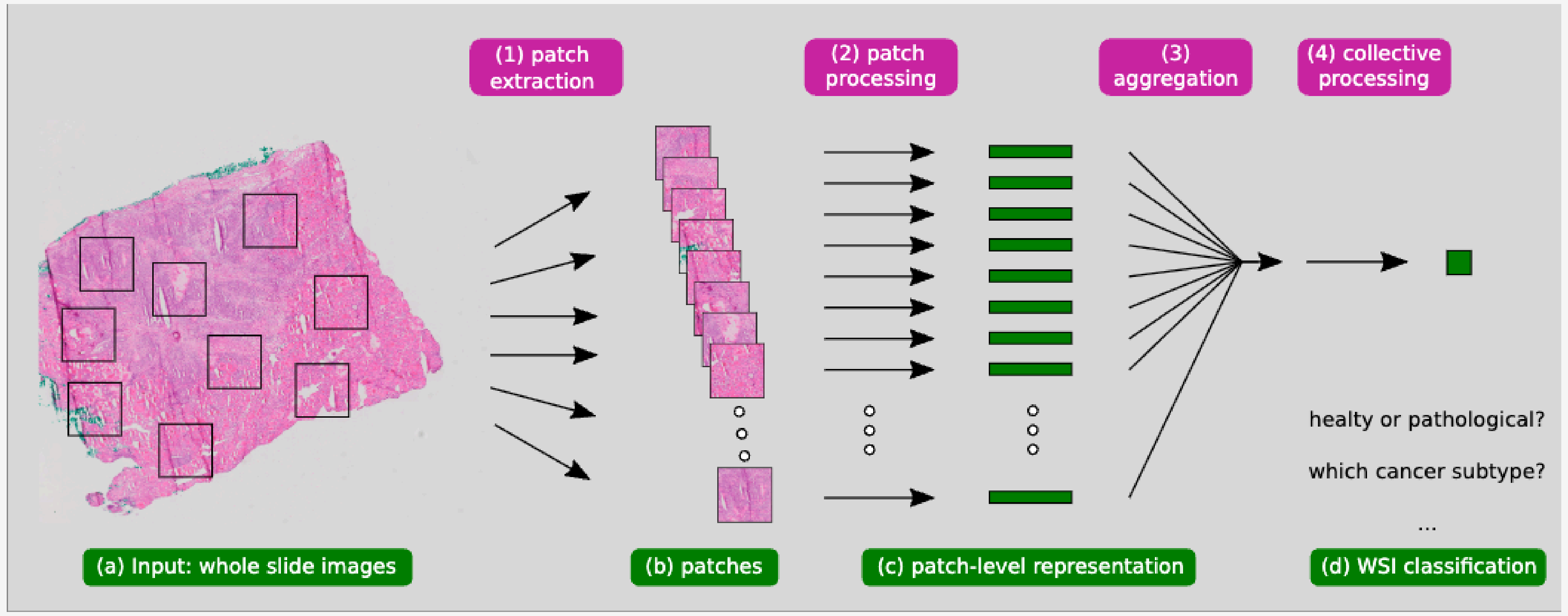
Methodological Approach (Part 2)

Multiple Instance Learning: Learning from Weak Labels

In digital pathology, detailed region annotations are rarely available due to the high cost and time required for expert labeling.

Weak supervision

Refers to using slide-level labels, where each whole slide image (WSI) is labeled for the overall diagnosis, but individual regions or patches are not annotated



Multiple Instance Learning (MIL)

Multiple Instance Learning (MIL) is a machine learning paradigm where data is organized into groups called "**bags**," each containing multiple "**instances**" (such as image patches). Unlike traditional supervised learning, labels are assigned only at the bag level, not to individual instances.

Multiple instance learning in digital pathology – Gadermayr & Tschuchnig, IEEE Transactions on Medical Imaging (2022)
MIL for whole slide image classification – Qu et al., CVPR (2023)

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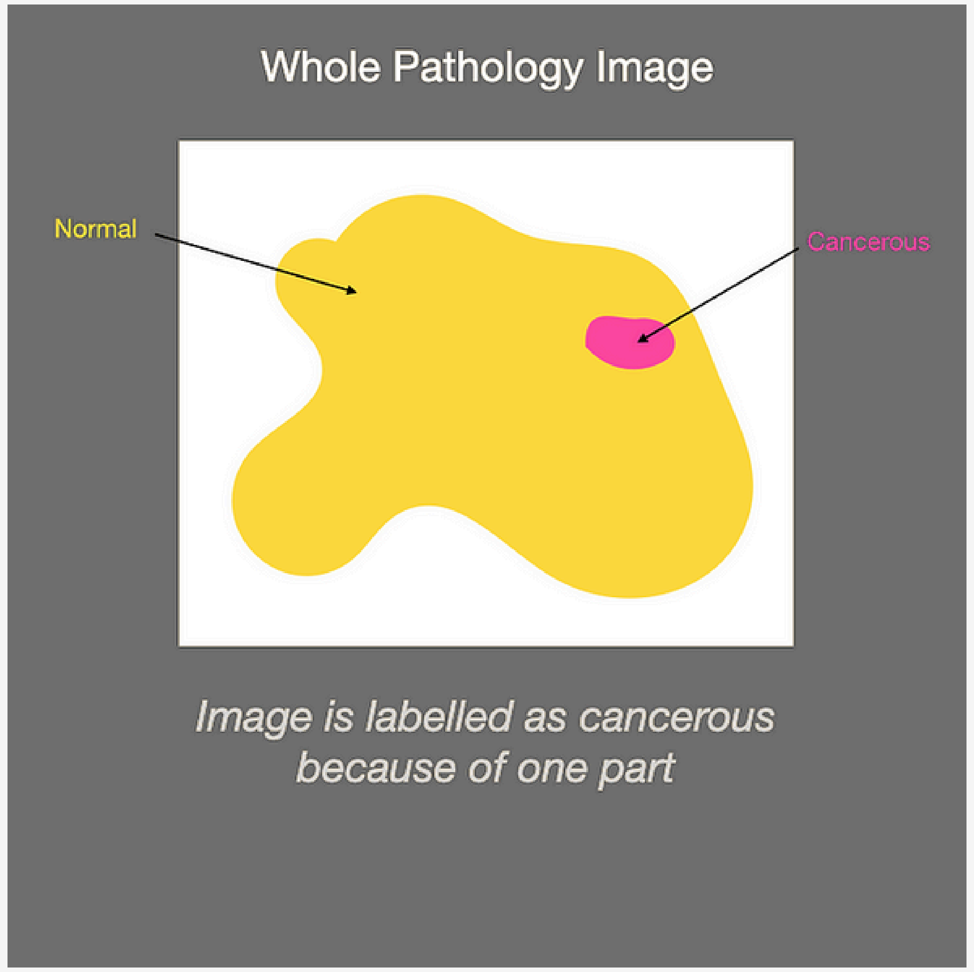
Methodological Approach (Part 3)

Why Use Multiple Instance Learning?



01 Motivation for MIL in Computational Pathology:

Traditional supervised learning requires exhaustive, pixel-level or region-level annotations, which are impractical for gigapixel WSIs. MIL leverages the fact that only slide-level labels are available, making it feasible to train powerful models even with limited annotation resources.

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02 MIL is especially effective for:

- Handling class imbalance and variability in tissue appearance.
- Focusing the model's attention on the most informative regions within each slide.
- Enabling scalable and reproducible analysis across large datasets.

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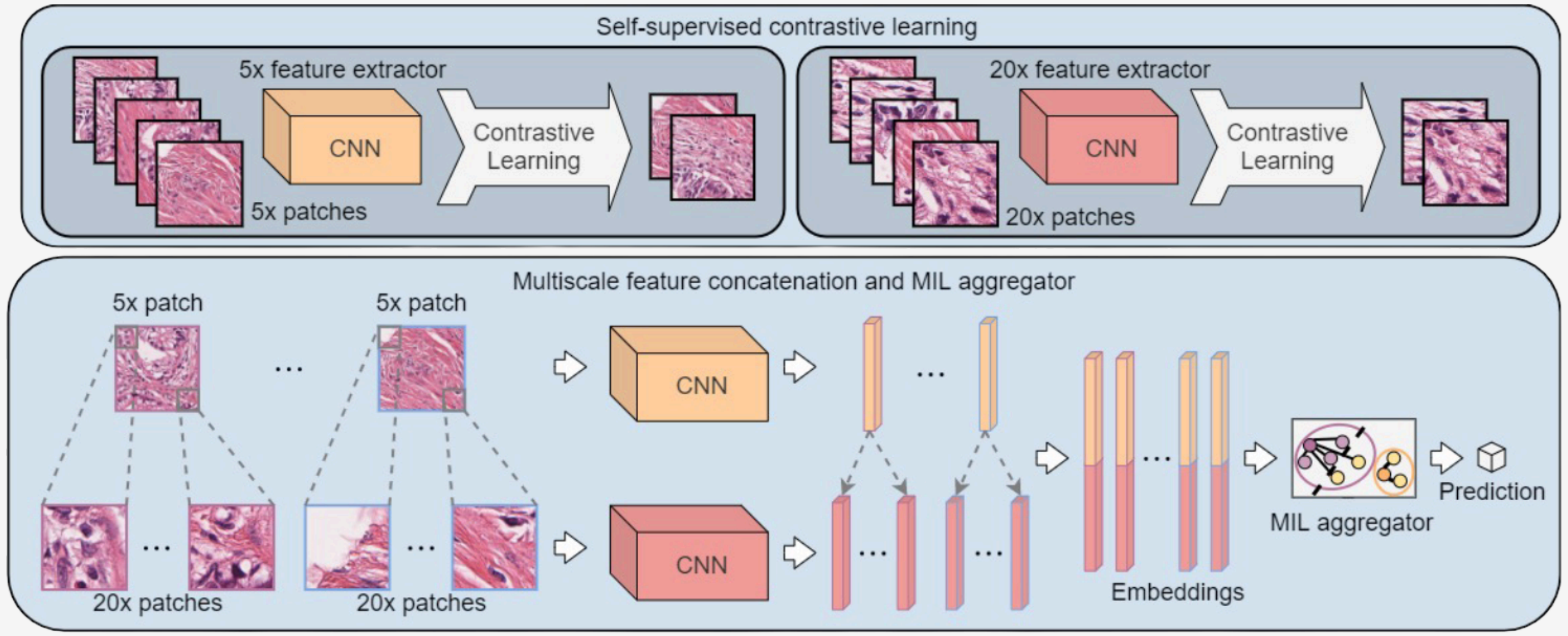
DSMIL: Dual-Stream Multiple Instance Learning

DSMIL introduces a novel dual-stream architecture for Multiple Instance Learning (MIL) in digital pathology, specifically designed to improve classification performance on whole slide images (WSI):



- **Separation of Instance-Level and Bag-Level Reasoning**
  DSMIL explicitly separates the process of identifying the most informative patch (instance-level) from aggregating contextual information across all patches (bag-level).
- **Max-Pooling and Contextual Attention**
  The first stream uses max-pooling to select the **single most critical patch** (the "critical instance"). The second stream computes **attention weights for all patches** based on their similarity to the critical instance, allowing the model to softly aggregate relevant contextual information from the entire slide.
- **Pyramidal Fusion (Multi-Magnification)**:
  DSMIL incorporates a **pyramidal fusion mechanism**, combining features extracted at multiple magnification levels.

Li, B., Li, Y., & Eliceiri, K. W. (2023). Dual-stream Multiple Instance Learning Network for Whole Slide Image Classification with Self-supervised Contrastive Learning. Medical Image Analysis

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DSMIL Results and Limitations

Key Results

- **Validation Performance:**
  - DSMIL achieved very high validation accuracy (up to 1.0) and validation AUC (up to 1.0) in several configurations
- **Test Performance:**
  - Despite excellent validation results, test accuracy consistently remained at 0.6, and test AUC values were low (ranging from 0.22 to 0.72), suggesting limited generalization to unseen data.
- **Variations in hyperparameters** (epochs, dropout, learning rate, weight decay) did not significantly affect test accuracy, which remained stable.
- **Multi-Scale Feature Fusion:**
  - The use of features extracted at multiple magnifications (5x and 10x) allowed DSMIL to integrate both fine and coarse tissue details.

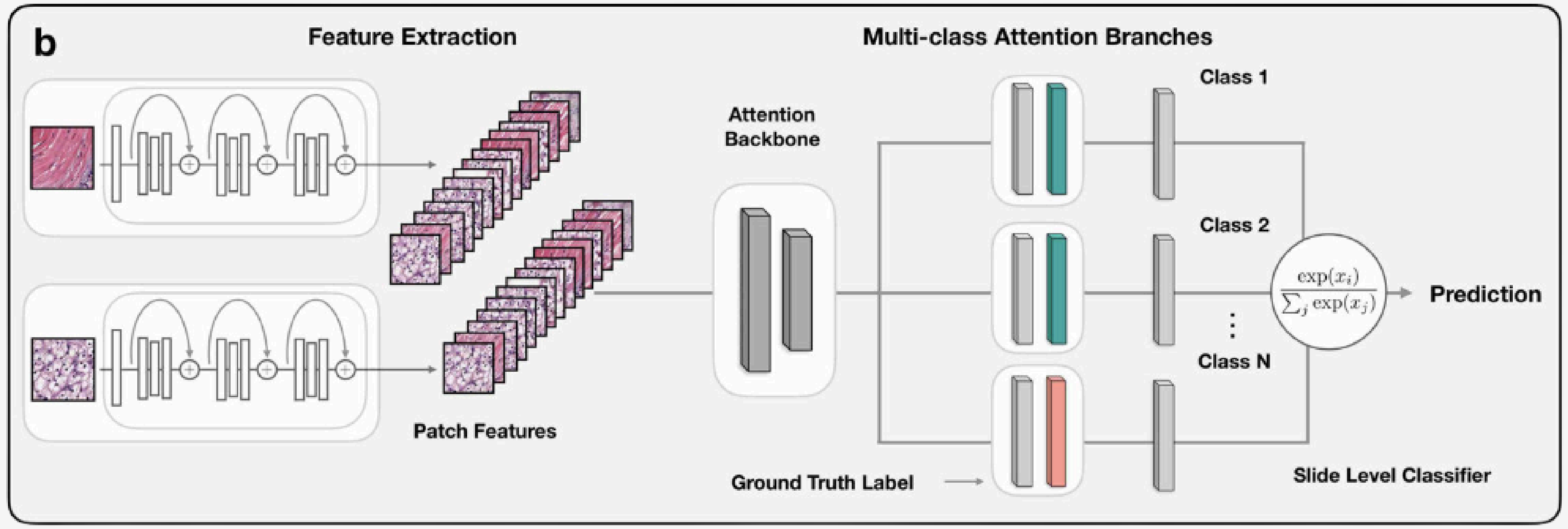
Epochs	Dropout	LR	WD	Test AUC	Val AUC	Test ACC	Val ACC
350	0.8	5e-5	1e-5	0.2778	1.0000	0.6	1.0
400	0.5	1e-5	1e-5	0.2222	1.0000	0.6	1.0
600	0.8	1e-4	1e-5	0.2778	1.0000	0.6	1.0
600	0.8	5e-5	1e-5	0.3333	1.0000	0.6	1.0
400	0.8	1e-5	1e-5	0.2222	1.0000	0.6	1.0
350	0.5	1e-5	1e-5	0.7222	0.9167	0.6	0.8
600	0.8	1e-5	1e-5	0.6111	0.9444	0.6	0.8
600	0.5	1e-4	1e-5	0.3889	0.9444	0.6	0.8
600	0.8	5e-5	1e-4	0.3333	0.9444	0.6	0.8
300	0.8	5e-5	1e-4	0.2778	1.0000	0.6	1.0

Main Limitations

- **Overfitting**:
  - The significant gap between validation and test performance suggests overfitting to the validation set.
- **Limited Test Discriminative Power:**
  - Low test AUC values indicate that DSMIL struggles to reliably distinguish between classes on the test set, despite high validation metrics.

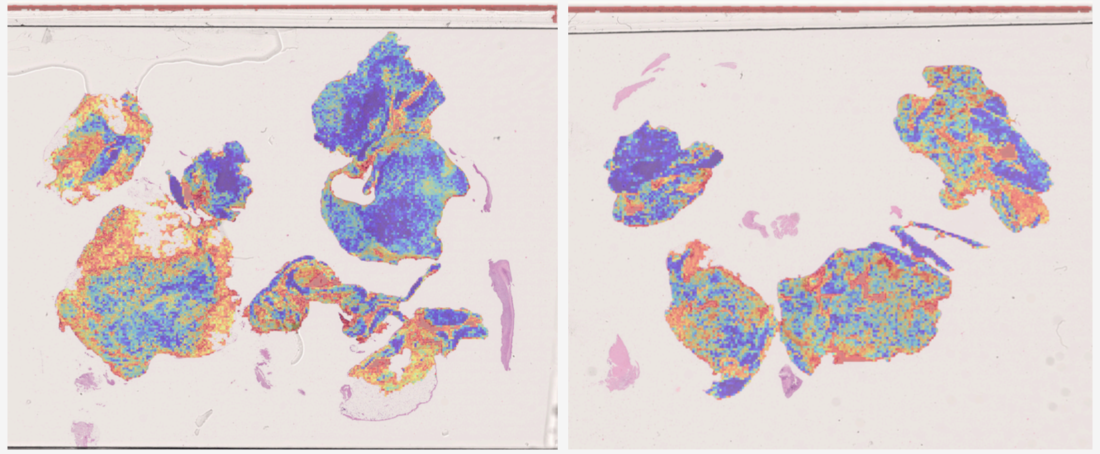
CLAM: Clustering-constrained Attention MIL

CLAM is an advanced deep learning framework designed for the classification of Whole Slide Images (WSI) in digital pathology, using only slide-level (weak) labels:



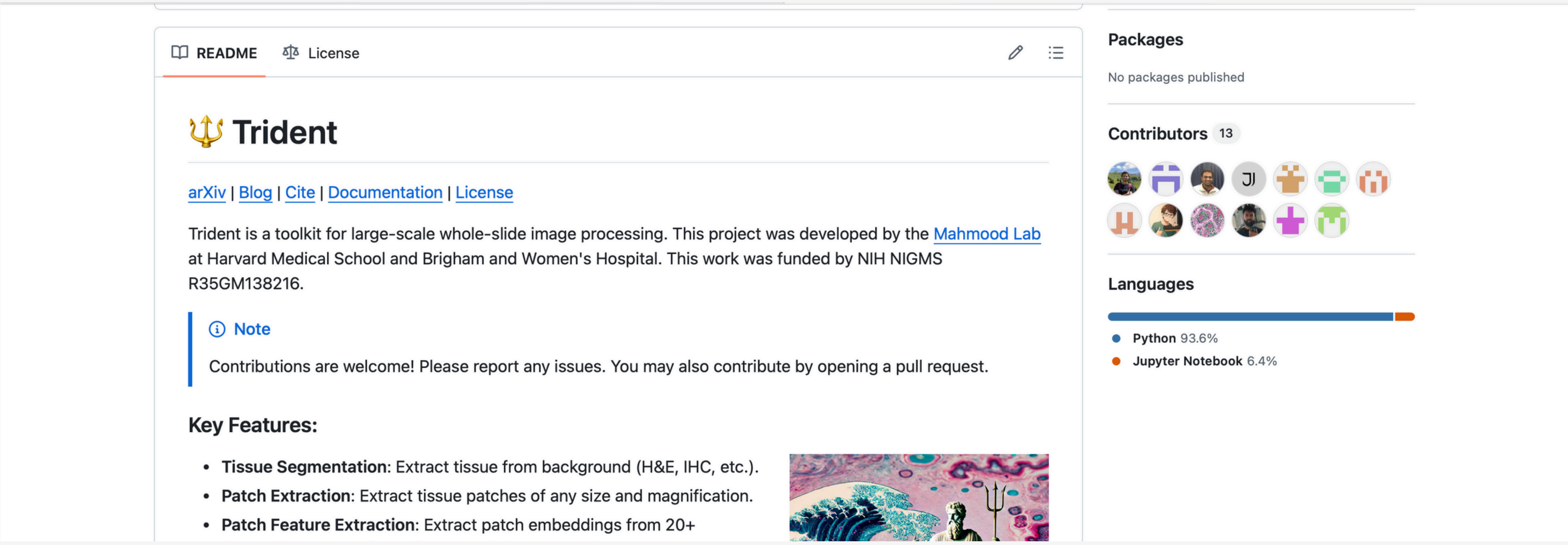
Data-efficient and weakly supervised computational pathology on WSI, Lu, Ming Y and Williamson,Nature Biomedical Engineering

- **Attention-based Pooling:**
  - Aggregates features from all image patches using an attention mechanism, allowing the model to automatically focus on the most diagnostically relevant regions within each slide.
- **Clustering of Relevant Patches:**
  - Implements instance-level clustering to distinguish between positive and negative evidence within each slide. Highly attended patches are grouped, refining the feature space and improving classification accuracy.
- **Interpretability via Attention Heatmaps:**
  - Generates attention heatmaps that highlight the regions most influential for the model's decision, providing visual interpretability and supporting clinical validation.



Trident: A Foundation for WSI Segmentation and Feature Extraction

An evolution of CLAM. Trident is a specialized Python framework developed for efficient segmentation and feature extraction from Whole Slide Images (WSI):



AI benchmarking for pathology – Zhang et al., Nature Communications (2023)

This **modular approach** ensures consistent and reproducible extraction of informative patch-level features for downstream machine learning tasks.

- **Robust Tissue Segmentation:**
  - Uses advanced models (e.g., DeepLabV3 pretrained on COCO) to accurately separate tissue from background, surpassing traditional thresholding methods and ensuring effective patch extraction across various stainings.
- **Patch Extraction:**
  - Divides the segmented tissue into manageable patches (e.g., 256×256 pixels), enabling scalable analysis of gigapixel WSIs on standard hardware.
- **Unified Feature Extraction with Supports multiple pretrained encoders, including:**
  - ResNet50 (ImageNet baseline)
  - UNI, UNIv2 (foundation models for pathology)
  - Phikon-v2 (biomarker prediction)
  - Other …

Data Augmentation

Why Perform Data Augmentation at the Feature Level Instead of the Slide Level?

Computational Efficiency

Whole Slide Images (WSIs) are gigapixel-sized and extremely resource-intensive to process. Traditional image-level augmentation (rotations, flips, color jittering) requires re-extracting features for every augmented image, which is computationally prohibitive for WSIs.

Scalability

By augmenting directly in the feature space (after patch extraction and encoding), new synthetic samples can be generated quickly without repeating the expensive segmentation and feature extraction steps.

Flexibility

Feature-level augmentation enables the use of advanced generative methods (e.g., extrapolation, diffusion models) that are not easily applicable at the raw image level.

Preservation of Semantics

Augmenting in feature space allows for controlled manipulation of the learned representations, ensuring that synthetic data remains meaningful and relevant for the classification task.

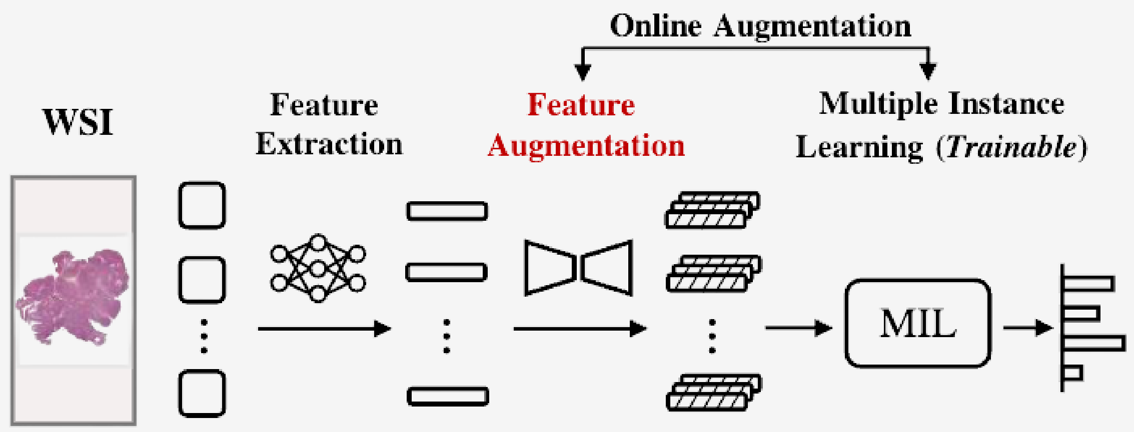
Feature-Level Data Augmentation

Extrapolation-Based Feature Augmentation

New synthetic samples are created by interpolating or extrapolating between existing feature vectors of **real patches.** New synthetic features are generated by extrapolating between pairs of real feature vectors:

$$\mathbf{c}' = (\mathbf{c}_j - \mathbf{c}_k)\lambda + \mathbf{c}_j$$

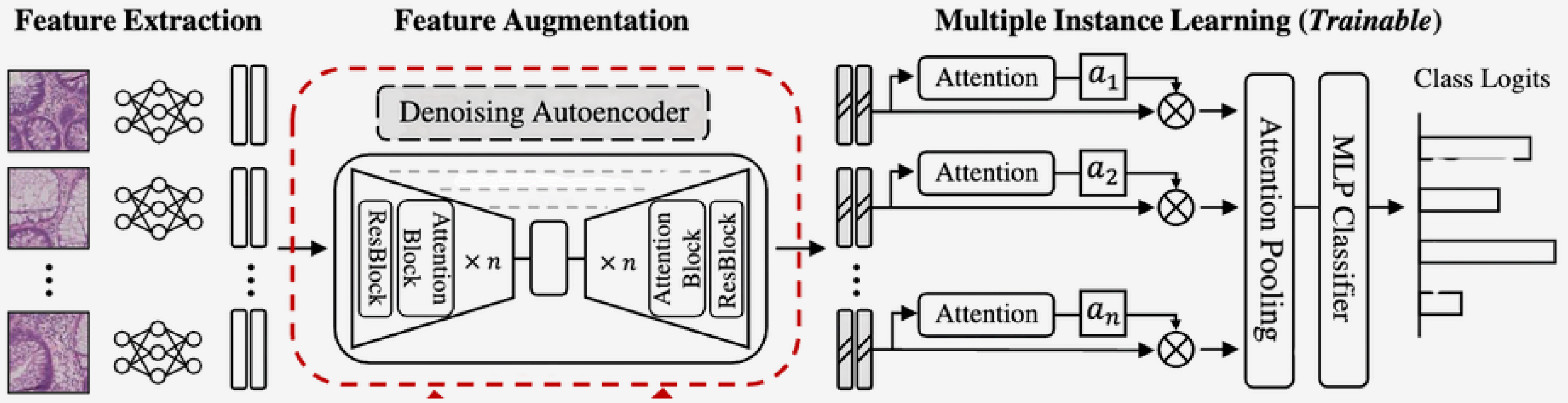
where λ controls the extrapolation.
This simple and efficient method increases **data diversity**, simulates new tissue variations, and helps the model **generalize better**, especially in small or imbalanced datasets. It has been shown to improve **robustness and reduce overfitting.**



Dataset augmentation in feature space – DeVries & Taylor, ICLR Workshop (2017)

AugDiff: Using Diffusion Models

AugDiff leverages generative models (Variational Autoencoders, U-Net, and diffusion models) to create realistic and diverse synthetic features directly in the learned feature space.
During augmentation, **random noise is added to latent rapresentation of feature vectors,** and the autoencoder denoises them to generate new synthetic samples that remain semantically consistent with the original tissue representation.
Compared to traditional augmentation methods, is particularly effective at maintaining the **biological meaning** of the features, supporting **online augmentation during model training,** and improving robustness and generalization.



AugDiff: diffusion-based feature augmentation – Shao et al., arXiv (2023)

Diverse Loss Functions:

Beyond standard cross-entropy, the study systematically tested:

- Cross Entropy Loss
- Focal Loss
- Weighted Cross Entropy (WCE)
- Supervised Contrastive Loss.

$$\mathcal{L}_{IWSCL}\left(q_{i,j}\right) = -\frac{1}{\left|F\left(q_{i,j}\right)\right|}\sum_{k_+\in F\left(q_{i,j}\right)} \log \frac{\exp\left({q_{i,j}}^{\top} k_+/\tau\right)}{\Sigma_{k^-\in F'\left(q_{i,j}\right)} \exp\left({q_{i,j}}^{\top} k_-/\tau\right)},$$

What is Contrastive Loss?

Supervised contrastive loss was introduced to make feature representations more robust and discriminative in weakly supervised whole slide image (WSI) classification. **Unlike standard cross-entropy**, which focuses only on individual predicted labels, contrastive loss shapes the feature space by pulling together samples from the same class and pushing apart those from different classes.

By applying contrastive loss to the most informative patches, the model becomes **more resilient** to class imbalance and limited annotations, ultimately improving classification

By applying contrastive loss to the most informative patches, the model becomes **more resilient** to class imbalance and limited annotations, ultimately improving classification

Baseline Results

Extractor	B.Loss	I.Loss	Test AUC	Val AUC	Test ACC	Val ACC
CLAM Extractor	CE	SVM	0.91	0.61	0.4	0.6
ResNet50 Trident	CE	SVM	0.44	0.44	0.4	0.2
Univ1 Trident	CE	SVM	0.83	0.27	0.6	0.4
Univ2 Trident	CE	SVM	0.55	0.77	0.4	0.6
Phikon Trident	CE	SVM	0.66	0.36	0.6	0.4

Table 1: Baseline with Cross Entropy Loss

Extractor	B.Loss	I.Loss	Test AUC	Val AUC	Test ACC	Val ACC
CLAM Extractor	Contr	SVM	0.91	0.66	0.4	0.4
ResNet50 Trident	Contr	SVM	0.58	0.44	0.4	0.4
Univ1 Trident	Contr	SVM	0.47	0.30	0.6	0.2
Univ2 Trident	Contr	SVM	0.41	0.77	0.4	0.4
Phikon Trident	Contr	SVM	0.44	0.33	0.6	0.4

Table 2: Baseline with Contrastive Loss

Extractor	B.Loss	I.Loss	Test AUC	Val AUC	Test ACC	Val ACC
CLAM Extractor	Focal	SVM	0.83	0.77	0.4	0.6
ResNet50 Trident	Focal	SVM	0.44	0.44	0.4	0.2
Univ1 Trident	Focal	SVM	0.77	0.41	0.6	0.4
Univ2 Trident	Focal	SVM	0.55	0.63	0.4	0.6
Phikon Trident	Focal	SVM	0.72	0.36	0.6	0.2

Table 3: Baseline with Focal Loss

Extractor	B.Loss	I.Loss	Test AUC	Val AUC	Test ACC	Val ACC
CLAM Extractor	WCE	SVM	0.91	0.61	0.4	0.6
ResNet50 Trident	WCE	SVM	0.44	0.44	0.4	0.2
Univ1 Trident	WCE	SVM	0.83	0.27	0.6	0.4
Univ2 Trident	WCE	SVM	0.55	0.77	0.4	0.6
Phikon Trident	WCE	SVM	0.66	0.36	0.6	0.4

Table 4: Baseline with Weighted CE Loss

The performance of different feature extractors was systematically evaluated to determine their impact on whole slide image (WSI) classification. Results show that the choice of feature extractor significantly influences both accuracy and discriminative ability (AUC), even when using the same model architecture and loss functions.

- CLAM Feature Extractor consistently achieved **the highest test AUC** (up to 0.91)
- Univ1 Trident and Phikon Trident sometimes matched the top test accuracy (0.6), but generally had **lower AUC** values
- ResNet50 Trident and Univ2 Trident displayed lower test AUC and accuracy
- Across all tested loss functions, the ranking of feature extractors remained stable, with **CLAM and Univ** extractors outperforming others.

Dimensionality Reduction and Model Performance

Extractor	B.Loss	I.Loss	Test AUC	Val AUC	Test ACC	Val ACC
CLAM Extractor	CE	SVM	0.69	0.66	0.6	0.6
ResNet50 Trident	CE	SVM	0.58	0.58	0.4	0.4
Univ1 Trident	CE	SVM	0.69	0.50	0.6	0.2
Univ2 Trident	CE	SVM	0.77	0.88	0.6	0.6
Phikon Trident	CE	SVM	0.50	0.58	0.2	0.2

Table 5: Baseline with Cross Entropy Loss + PCA

Extractor	B.Loss	I.Loss	Test AUC	Val AUC	Test ACC	Val ACC
CLAM Extractor	Contrastive	SVM	0.83	0.36	0.6	0.4
ResNet50 Trident	Contrastive	SVM	0.50	0.70	0.6	0.4
Univ1 Trident	Contrastive	SVM	0.69	0.50	0.6	0.4
Univ2 Trident	Contrastive	SVM	0.83	0.75	0.4	0.6
Phikon Trident	Contrastive	SVM	0.72	0.58	0.6	0.2

Table 6: Baseline with Contrastive Loss + PCA

Extractor	B.Loss	I.Loss	Test AUC	Val AUC	Test ACC	Val ACC
CLAM Extractor	Focal	SVM	0.83	0.44	0.6	0.4
ResNet50 Trident	Focal	SVM	0.58	0.77	0.6	0.4
Univ1 Trident	Focal	SVM	0.69	0.50	0.6	0.4
Univ2 Trident	Focal	SVM	0.88	0.75	0.6	0.8
Phikon Trident	Focal	SVM	0.77	0.58	0.8	0.2

Table 7: Baseline with Focal Loss + PCA

Extractor	B.Loss	I.Loss	Test AUC	Val AUC	Test ACC	Val ACC
CLAM Extractor	WCE	SVM	0.83	0.36	0.6	0.4
ResNet50 Trident	WCE	SVM	0.50	0.78	0.6	0.4
Univ1 Trident	WCE	SVM	0.70	0.50	0.6	0.4
Univ2 Trident	WCE	SVM	0.83	0.75	0.4	0.6
Phikon Trident	WCE	SVM	0.72	0.58	0.6	0.2

Table 8: Baseline with Weighted CE Loss + PCA

Principal Component Analysis (PCA), preserving the of **99% variance**, of was applied to reduce the dimensionality of feature vectors extracted from whole slide images (WSI), aiming to streamline **computation** and potentially improve **generalization**.

The results indicate that PCA can maintain:

- **Performance Trends**, With PCA, feature extractors like **Univ2 Trident and CLAM** often achieved the highest test and validation AUC, demonstrating strong discriminative capability and generalization.
- **Loss Function Compatibility**, Across all tested loss functions the ranking of feature extractors remained consistent, with PCA not fundamentally altering which models performed best.
- **Key Insight,** the choice of feature extractor remains the most critical factor for WSI classification performance. PCA serves as a useful tool for reducing computational complexity.

Impact of Extrapolation on Classification

Extractor	B.Loss	I.Loss	PCA	Test AUC	Val AUC	Test ACC	Val ACC
Aug Resnet50	CE	SVM	No	0.91	0.89	0.90	0.90
Aug Univ1	CE	SVM	No	1.00	1.00	0.90	0.90
Aug Univ2	CE	SVM	No	1.00	1.00	1.00	1.00
Aug Phikon	CE	SVM	No	0.98	0.98	0.80	0.90
Aug Resnet50	CE	SVM	Si	0.98	0.98	0.80	0.80
Aug Univ1	CE	SVM	Si	1.00	1.00	0.90	0.90
Aug Univ2	CE	SVM	Si	1.00	1.00	1.00	1.00
Aug Phikon	CE	SVM	Si	0.97	0.85	0.80	0.80

Table 9: CE + Data Aug (Ext), w/o PCA

Extractor	B.Loss	I.Loss	PCA	Test AUC	Val AUC	Test ACC	Val ACC
Aug Resnet50	Contrastive	SVM	No	0.94	0.88	0.90	0.90
Aug Univ1	Contrastive	SVM	No	1.00	1.00	1.00	1.00
Aug Univ2	Contrastive	SVM	No	1.00	1.00	0.88	0.88
Aug Phikon	Contrastive	SVM	No	0.99	0.96	0.90	0.90
Aug Resnet50	Contrastive	SVM	Si	0.98	0.97	0.80	0.80
Aug Univ1	Contrastive	SVM	Si	0.97	0.97	0.80	0.80
Aug Univ2	Contrastive	SVM	Si	1.00	1.00	1.00	1.00
Aug Phikon	Contrastive	SVM	Si	0.97	0.90	0.80	0.80

Table 10: Contrastive + Data Aug (Ext), w/o PCA

Extractor	B.Loss	I.Loss	PCA	Test AUC	Val AUC	Test ACC	Val ACC
Aug Resnet50	Focal	SVM	No	0.91	0.89	0.90	0.90
Aug Univ1	Focal	SVM	No	1.00	1.00	0.90	0.90
Aug Univ2	Focal	SVM	No	1.00	1.00	1.00	1.00
Aug Phikon	Focal	SVM	No	0.98	1.00	0.90	0.90
Aug Resnet50	Focal	SVM	Si	0.98	0.97	0.80	0.80
Aug Univ1	Focal	SVM	Si	0.98	1.00	0.90	1.00
Aug Univ2	Focal	SVM	Si	1.00	1.00	1.00	1.00
Aug Phikon	Focal	SVM	Si	0.98	0.88	0.80	0.80

Table 11: Focal + Data Aug (Ext), w/o PCA

Feature Extractor	B.Loss	I.Loss	PCA	Test AUC	Val AUC	Test ACC	Val ACC
Aug Resnet50	WCE	SVM	No	0.91	0.89	0.90	0.90
Aug Univ1	WCE	SVM	No	1.00	1.00	0.90	0.90
Aug Univ2	WCE	SVM	No	1.00	1.00	1.00	1.00
Aug Phikon	WCE	SVM	No	0.98	0.98	0.80	0.90
Aug Resnet50	WCE	SVM	Si	0.98	0.97	0.80	0.80
Aug Univ1	WCE	SVM	Si	0.97	0.97	0.80	0.80
Aug Univ2	WCE	SVM	Si	1.00	1.00	1.00	1.00
Aug Phikon	WCE	SVM	Si	0.97	0.90	0.80	0.80

Table 12: WCE + Data Aug (Ext), w/o PCA

Extrapolation-Based Augmentation

- This approach produced dramatic gains: models like **Aug Univ2 Trident** achieved perfect test and validation AUC and accuracy (1.00), regardless of whether PCA was used.
- Augmented versions of **ResNet50, UNI, and Phikon-v2 consistently** reached high AUC (⩾0.90) and accuracy (⩾0.80), confirming that both general-purpose and pathology-specific extractors benefit from this method.
- The use of PCA alongside augmentation did **not reduce performance** and sometimes slightly improved results, demonstrating that dimensionality reduction is fully compatible with feature-level augmentation.

Impact of AugDiff on Classification

Extractor	B.Loss	I.Loss	PCA	Test AUC	Val AUC	Test ACC	Val ACC
AugDiff R50	CE	SVM	No	0.44	0.66	0.22	0.55
AugDiff R50	CE	SVM	Si	0.50	0.67	0.22	0.33
AugDiff R50	Contr	SVM	No	0.40	0.61	0.22	0.55
AugDiff R50	Contr	SVM	Si	0.52	0.71	0.22	0.44
AugDiff R50	Focal	SVM	No	0.44	0.62	0.22	0.55
AugDiff R50	Focal	SVM	Si	0.50	0.67	0.22	0.33
AugDiff R50	WCE	SVM	No	0.40	0.62	0.22	0.55
AugDiff R50	WCE	SVM	Si	0.50	0.67	0.22	0.33

Table 13: all Losses + AugDiff w/o PCA

AugDiff Diffusion Model Augmentation

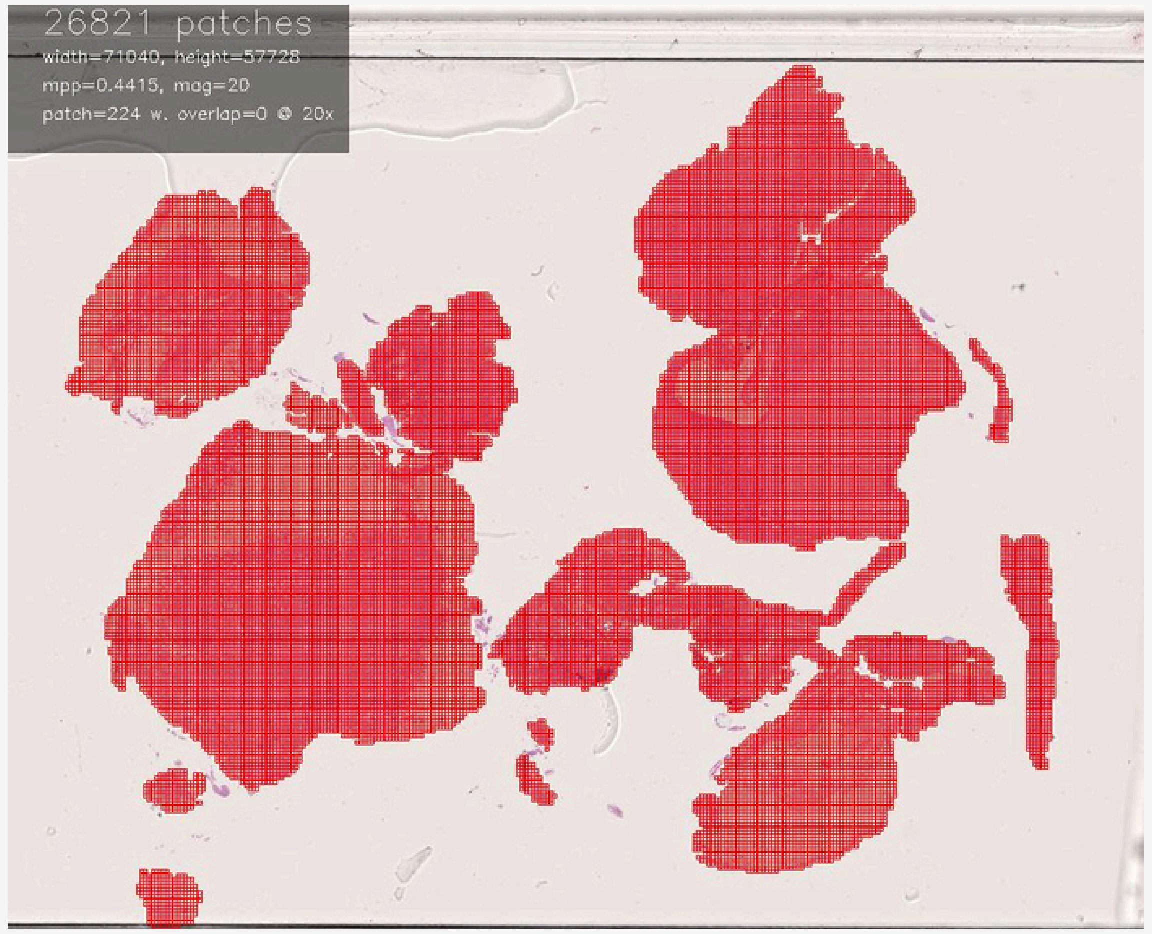
- This method, tested on ResNet50 Trident, **did not reach the performance levels of extrapolation-based augmentation**. Test AUC values ranged from 0.40 to 0.52, and test accuracy remained at 0.22.
- The limited improvement may be due to factors such as low feature dimensionality, a reduced number of diffusion steps, or subsampling in the diffusion process, which could restrict the diversity and realism of generated synthetic features.

Conclusion

The project demonstrated that weakly supervised and semi-supervised learning strategies are highly effective for classifying malignant pleural mesothelioma

- **The choice of feature extractor** was the most influential factor for model performance
- **Advanced loss functions** (Focal, Weighted Cross Entropy, Supervised Contrastive Loss) further improved robustness, particularly in the presence of class imbalance.
- Feature-level **data augmentation** was transformative
- **Interpretability tools** such as blockmaps provided valuable insights into model decisions

Overall, the **UNIv2 Trident** Feature Extractor archives exceptional **robustness and generalization** in combination with Data Augmentation.



Future Works

- Larger and More Diverse Datasets
- Advanced Self-Supervised Learning
- Domain Adaptation
- Model Optimization and Regularization
- Clinical Integration and Validation
- Enhanced Interpretability

THANK YOU

Q&A